

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer with 5+ years building real-time, high-performance Node.js and TypeScript systems on Google Cloud and AWS. Led a zero-downtime migration of 20+ production apps from MongoDB to Firestore via Google Pub/Sub at MTailor, ran 2M-record analytical jobs against Firestore at Dr Wealth, and architected Kafka + NATS + Kubernetes microservices for 4 NGO tenants at CodeBits. Strong fit for scalable Node/TS backend platforms where reliability, observability and clean APIs matter.

CORE COMPETENCIES

Node.js + TypeScript

MongoDB + SQL

Google Cloud (GCP)

Pub/Sub + Kafka

Kubernetes + Docker

Real-time data sync

High-performance APIs

Distributed systems

WORK EXPERIENCE

MTailor

September 2022 – Present

Software Engineer (US Remote) — reporting to CTO

- Led **zero-downtime migration of 20+ production Node.js apps** from Parse/MongoDB to Firebase/GCP, owning architecture through cutover
- Built **real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub** — idempotent, high-throughput event-driven message processing during live migration
- Implemented a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion
- Migrated file storage from Amazon S3 to GCS via a Python script with retries and dry-run mode
- Implemented Express Shipping integration, generating **\$40 additional revenue per order**
- Saved the company **\$5,000/month** by migrating all services off AWS

MTailor

March 2022 – July 2022

Contractor (US Remote) — Upwork-sourced backend migration

- Built a Node.js migration tool to read MongoDB and write to Firestore after complex transformation logic
- Wrote documentation for fellow engineers on working with the new Firebase SDKs (Node.js and Python)
- Trained the Ops team on the new Firebase Dashboard and backup procedures

Dr Wealth

August 2021 – January 2022

Software Engineer (Singapore Remote)

- Optimized a PWA delivering real-time stock prices from a Firebase backend
- Built ad-hoc jobs to **query over 2 million Firestore records**, keeping customer prices current from Morningstar APIs
- Extended backend APIs using **Firebase Cloud Functions and Express.js** hosted on Heroku
- Added responsive UI pages to the PWA using HTML, Tailwind, and React

Mind2matter

January 2022 – July 2022

Software Engineer (US Remote) — reporting to CTO

- Built backends for DeFi applications using Web3 and Node.js
- Delivered React UIs from Figma designs under tight timelines

CodeBits

January 2020 – July 2021

Software Engineer / Lead (Uganda Remote)

- Led a team of 4 developers building legal-tech systems for NGOs in Uganda
- Architected a microservices backend using Apache Kafka, Docker, Kubernetes, NATS Streaming, and gRPC** for FIDA Uganda's multi-tenant case management platform
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda
- Developed a cross-platform mobile app for LASPNET using React Native and Expo
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC

EDUCATION

B.Sc. Computer Engineering · [Makerere University, Kampala](#)

2018 – 2022

SKILLS

Proficient: Node.js, TypeScript, JavaScript, MongoDB, SQL, GCP (Firestore, Pub/Sub, Cloud Functions, GCS), Python, Firebase, React, Git, Linux

Intermediate: Go, AWS (S3, EC2, EBS), Rust, React Native, Redis (familiar), Kubernetes, Docker, Kafka, NATS, gRPC

Languages: English (native), Kiswahili